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DS PROGRAMMING

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Fibonacci series (N ≤ 50)

FIBONACCI SERIES (N ≤ 50)

Develop a C program to print the first N terms of the Fibonacci series (N ≤ 50) without using recursion. Use two variables to generate the series iteratively and store the series in
an array. The program should also calculate and display the sum of the series and identify the largest term. Input validation is mandatory for N (1 ≤ N ≤ 50).
Sample Input/Output: text
Enter number of terms (1-50): 10
Fibonacci Series: 0 1 1 2 3 5 8 13 21 34
Sum of series: 88
Largest term: 34
Enter number of terms (1-50): 60
Invalid input! Please enter terms between 1 and 50.

PROGRAM EDITOR

C

Run Save

```c
#include <stdio.h>

int main() {
    int n, i, f[50];

    scanf("%d", &n);

    if (n < 1 || n > 50) {
        printf("Invalid input! Please enter terms between 1 and 50.");
        return 0;
    }

    f[0] = 0;
    if (n > 1)
        f[1] = 1;

    for (i = 2; i < n; i++)
        f[i] = f[i - 1] + f[i - 2];

    int sum = 0, largest;
```

OUTPUT

```
Fibonacci Series: 0 1 1 2 3 5 8 13 21 34
Sum of series: 88
Largest term: 34
```

INPUT

10